



Submitted by

Name: Muhammad Abdul Kareem

Roll no: BSEF23M014

Semester: 3rd

Section: BS-SE(Morning)

Lab Report: 1st

Subject: Applied Physic

Submitted to

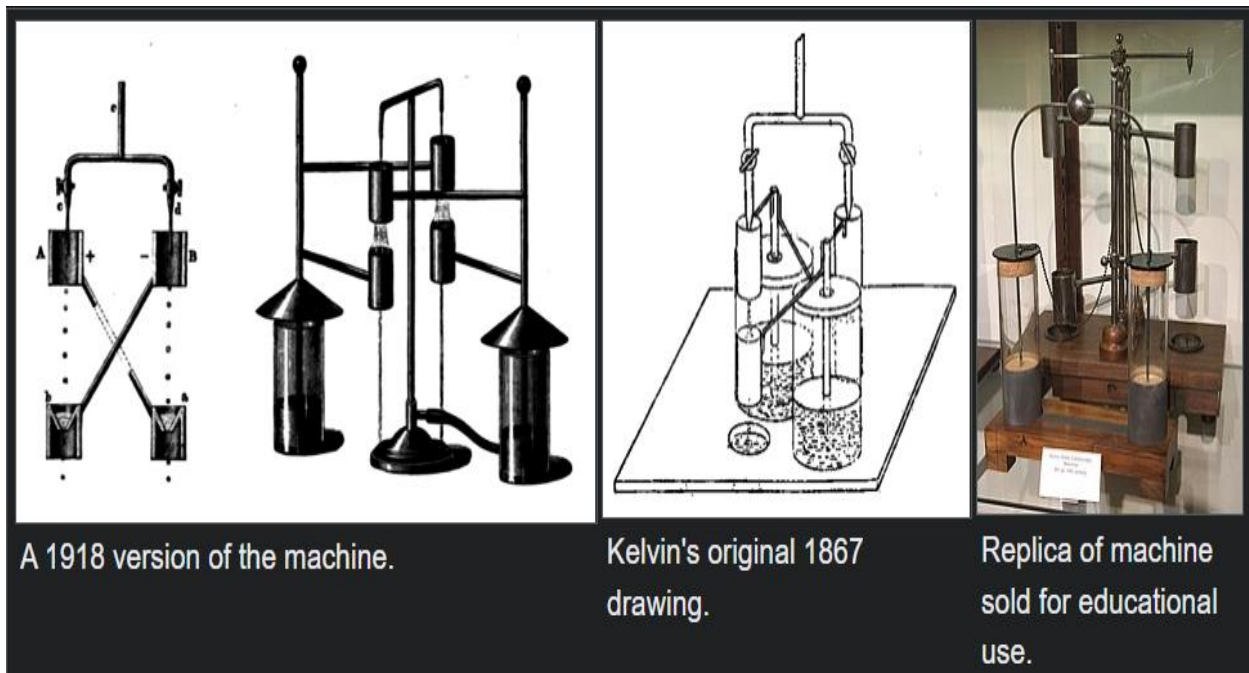
Teacher: Khizar

LAB REPORT

VAN de Graaff generator

➤ **Background:**

The concept of an electrostatic generator, where charge is mechanically transported in small amounts to the interior of a high-voltage electrode, traces back to the Kelvin water dropper, invented in 1867 by William Thomson (Lord Kelvin). In this device, charged water droplets fall into a bucket carrying the same charge polarity, gradually increasing the charge. Here, gravity moves the droplets against the opposing electrostatic field. Lord Kelvin first suggested using a belt to carry charge instead of water. Its picture is given below.



Augusto Righi built the earliest electrostatic machine employing a continuous belt in 1872. Righi's design used an Indian rubber belt with wire rings as charge carriers, which moved into a spherical metal electrode. The charge was transferred to the belt from the grounded lower roller via electrostatic induction from a charged plate. Around 1890, John Gray also invented a similar belt-driven machine. In 1903, Juan Burboa introduced a more complex belt machine.

A significant influence on Van de Graaff's work was a generator developed in the 1920s by W.F. G. Swann, where the charge was carried by falling metal balls, drawing again from the principles of the Kelvin water dropper.

➤ Introduction:

The Van de Graaff generator, invented in 1929 by American physicist Robert J. Van de Graaff, is an electrostatic device that uses a moving belt to transport electric charge to a hollow metal sphere, building up very high voltages. It was inspired by earlier concepts like the “Kelvin water dropper of 1867”, which created strong electric fields by continuously accumulating charge. With the help of his colleague Nicholas Burke, He demonstrated the first model in October 1929, achieving around 1.5 million volts—potentials that increased to 2.5 million volts when enclosed in insulating gases like sulfur hexafluoride (SF₆) to prevent electrical discharges.

➤ Purpose:

Originally it was developed as a particle accelerator for nuclear physics, the generator’s high voltages could accelerate subatomic particles in an evacuated tube, providing critical insights into atomic energy and matter structure. This technology, the most powerful of its kind until the invention of the cyclotron in the 1930s, continues to be used for generating particle and X-ray beams in nuclear research and medicine. Open-air Van de Graaff generators can produce up to 5 megavolts, though modern enclosed versions in pressurized gas tanks achieve voltages as high as 25 megavolts. Small versions of the generator, often producing around 100 kilovolts, are popular in educational and museum settings for demonstrations of electrostatic principles.

➤ Principle:

The Van de Graaff generator operates on the principle of “electrostatic induction”, where a moving belt transports electric charge to a metal sphere, resulting in high voltage accumulation and two other phenomena:

- Charge accumulation on a hollow conductor: Any charge given to a hollow conductor is distributed uniformly on its outer surface.
- Electrical discharge at sharp points: Charged particles are readily discharged at pointed conductors.

➤ Construction:

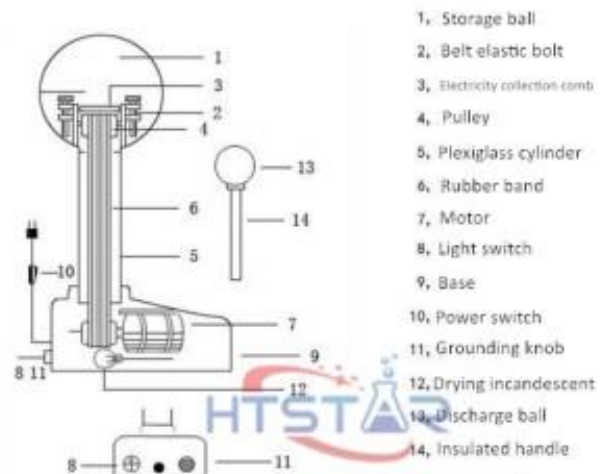
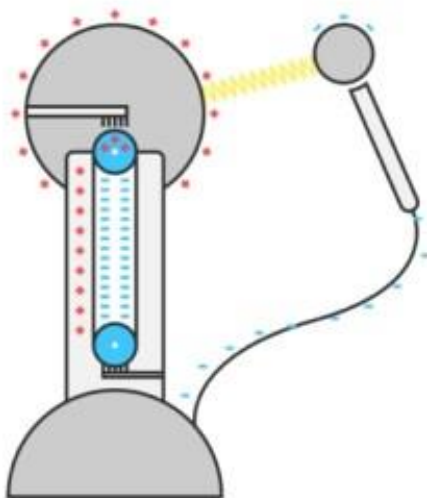
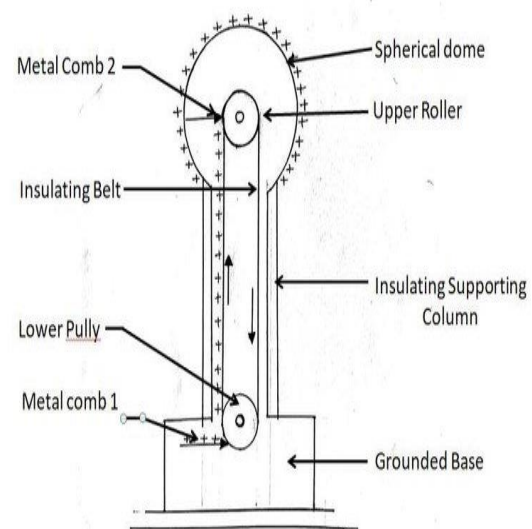
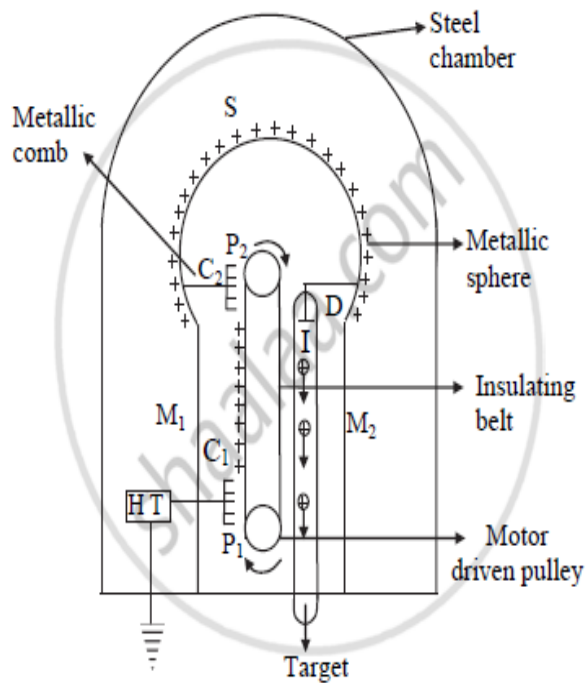
The construction of a Van de Graaff generator involves several key components that work together to generate high-voltage static electricity. These components are given below:

1. Base.
2. Motor.
3. Rubber belt
4. 2-Rollers

5. Spray comb
6. Insulating rod
7. Receiving comb
8. Hollo sphere
9. Power supply
10. Discharge sphere (optional)

➤ **Diagram:**

The diagram of the Van De Graaff Generator is given below:



➤ **Working:**

The Van de Graaff generator generates high voltage through a series of processes involving a moving belt:

1. **Charge Generation:** The rubber belt travels over two rollers made from different materials, acquiring electric charge from the spray comb through the triboelectric effect.
2. **Charge Transport:** The charged belt moves upward along an insulated rod, carrying the accumulated charge to the generator's receiving comb.
3. **Charge Transfer:** Metal combs with sharp points are positioned near the rollers. The lower comb deposits charge onto the belt, while the upper comb removes charge from the belt and transfers it to a large metal sphere.
4. **High Voltage Accumulation:** The charge builds up on the metal sphere, creating a very high voltage, often reaching several megavolts, resulting in strong electric fields.
5. **Discharge:** Once the voltage is sufficiently high, it can cause visible sparks or arcs to jump to nearby conductive objects.

This mechanism allows the Van de Graaff generator to produce high-voltage direct current (DC) electricity, making it valuable for educational demonstrations and research applications.

➤ **Applications:**

The Van de Graaff generator has a range of applications due to its ability to generate high-voltage electricity at low currents. Here are its primary uses:

1. **Educational Demonstrations:** Small Van de Graaff generators are widely used in classrooms and science museums to demonstrate principles of static electricity, including charge transfer, electric fields, and high-voltage discharges (sparks). They visually illustrate fundamental concepts in electrostatics.
2. **Particle Accelerators:** Originally designed as a particle accelerator, the Van de Graaff generator boosts particles to high energies. This application is still relevant in nuclear physics research, where generators are often housed in pressurized tanks with insulating gases to prevent electrical discharge and enhance performance.
3. **Nuclear Research:** In scientific research, Van de Graaff generators generate particle and X-ray beams. These high-energy beams are used to study nuclear

reactions and atomic nuclei, contributing to advances in fields like particle physics and radiation research.

4. **Medical Applications:** In medical settings, Van de Graaff generators are valuable for nuclear medicine and radiation therapy. They generate powerful X-ray beams for imaging and treatment, supporting therapeutic applications in cancer treatment and other medical procedures.
5. **Industrial Testing and Calibration:** High-voltage testing is essential in electrical engineering. Van de Graaff generators are used to test the insulation of materials, calibrate equipment, and simulate high-voltage scenarios for industrial applications.